

AREA — Short Revision Notes

Remember Level: Recall Facts & Basic Knowledge



Square



Rectangle



Triangle



Parallelogram



Trapezium



Circle

1 Key Definitions

Term	Definition	Example
Area	The amount of space occupied by a surface	The space inside a shape
Square Unit	Unit used to measure area	cm ² , m ² , km ²
Base	The side of a shape used as the bottom/reference	Bottom side of a triangle
Height / Perpendicular Height	The perpendicular distance from base to the opposite side	Measured at 90° to the base
Parallelogram	A quadrilateral with two pairs of parallel sides	Opposite sides are equal and parallel
Trapezium	A quadrilateral with one pair of parallel sides	Only one pair of opposite sides are parallel
Radius (r)	Distance from center to any point on a circle	Half the diameter
Diameter (d)	Distance across a circle through the center	Twice the radius
π (Pi)	A constant value	22/7 or 3.142

2 Area Formulas

Shape	Formula	Variables
Square	$A = a^2$	a = side length
Rectangle	$A = l \times b$	l = length, b = breadth
Triangle	$A = \frac{1}{2} \times a \times h$	a = base, h = perpendicular height
Parallelogram	$A = a \times h$	a = base, h = perpendicular height
Trapezium	$A = \frac{1}{2} \times (a + b) \times h$	a, b = parallel sides, h = perpendicular height
Circle	$A = \pi r^2$	r = radius

3 Quick Reference Table

Shape	Formula	Variables
Square	a^2	a = side
Rectangle	$l \times b$	l = length, b = breadth
Triangle	$\frac{1}{2} \times \text{base} \times \text{height}$	base, perpendicular height
Parallelogram	$\text{base} \times \text{height}$	base, perpendicular height
Trapezium	$\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$	parallel sides, height
Circle	πr^2	r = radius

4 Key Symbols

Symbol	Meaning	Used For
A	Area	All shapes
a	Side length	Square, Triangle base
l	Length	Rectangle
b	Breadth / Base	Rectangle, Triangle, Parallelogram
h	Height	Triangle, Parallelogram, Trapezium
r	Radius	Circle
d	Diameter	Circle
π	Pi = $22/7 \approx 3.142$	Circle
cm ²	Square centimetres	Area unit
m ²	Square metres	Area unit

5 Key Relationships

Rectangle to Triangle

Shape	Relationship
Triangle	Area = $\frac{1}{2} \times \text{base} \times \text{height}$
Rectangle	Area = base $\times$ height
Triangle	Triangle is half of a rectangle with the same base and height

Parallelogram to Rectangle

Shape	Relationship
Parallelogram	Area = base $\times$ height
Rectangle	Area = base $\times$ height
Relationship	Parallelogram and rectangle with the same base and height have equal area

Circle to Square

Shape	Relationship
Circle	Area = πr^2
Square	Area = a^2
Relationship	Circle inscribed in square: diameter = side of square

6 Check Yourself — Quick Questions

No.	Question	Answer
1	What is the formula for area of a square?	a^2
2	What is the formula for area of a rectangle?	$l \times b$
3	What is the formula for area of a triangle?	$\frac{1}{2} \times \text{base} \times \text{height}$
4	What is the formula for area of a parallelogram?	$\text{base} \times \text{height}$
5	What is the formula for area of a trapezium?	$\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
6	What is the formula for area of a circle?	πr^2
7	What is the value of π ?	22/7 or 3.142
8	What is the unit of area?	cm^2 , m^2 , km^2
9	What is the height of a triangle?	Perpendicular distance from base to opposite vertex
10	How many parallel sides does a trapezium have?	One pair
11	How many parallel sides does a parallelogram have?	Two pairs
12	What is the area of a square of side 5 cm?	25 cm^2
13	What is the area of a rectangle with length 10 cm and breadth 6 cm?	60 cm^2
14	What is the area of a triangle with base 12 cm and height 5 cm?	30 cm^2
15	What is the area of a parallelogram with base 10 cm and height 5 cm?	50 cm^2

7 Quick Memory Aids

AREA FORMULAS:

SQUARE:	$A = a^2$
RECTANGLE:	$A = l \times b$
TRIANGLE:	$A = \frac{1}{2} \times b \times h$
PARALLELOGRAM:	$A = b \times h$
TRAPEZIUM:	$A = \frac{1}{2} \times (a+b) \times h$
CIRCLE:	$A = \pi r^2$

MEMORY TRICKS:

- Square: "Side squared"
- Rectangle: "Length $\times$ Breadth"
- Triangle: "Half base $\times$ height"
- Parallelogram: "Base $\times$ height"
- Trapezium: "Half sum of parallel sides $\times$ height"
- Circle: " π r squared"

KEY UNITS:

Area is ALWAYS in square units:

cm^2 , m^2 , km^2 , mm^2

$$1 \text{ m}^2 = 10,000 \text{ cm}^2$$

$$1 \text{ km}^2 = 1,000,000 \text{ m}^2$$

Revision Tip: Memorize all area formulas — they are the foundation for all area problems! Use the memory tricks to help you remember each formula correctly.